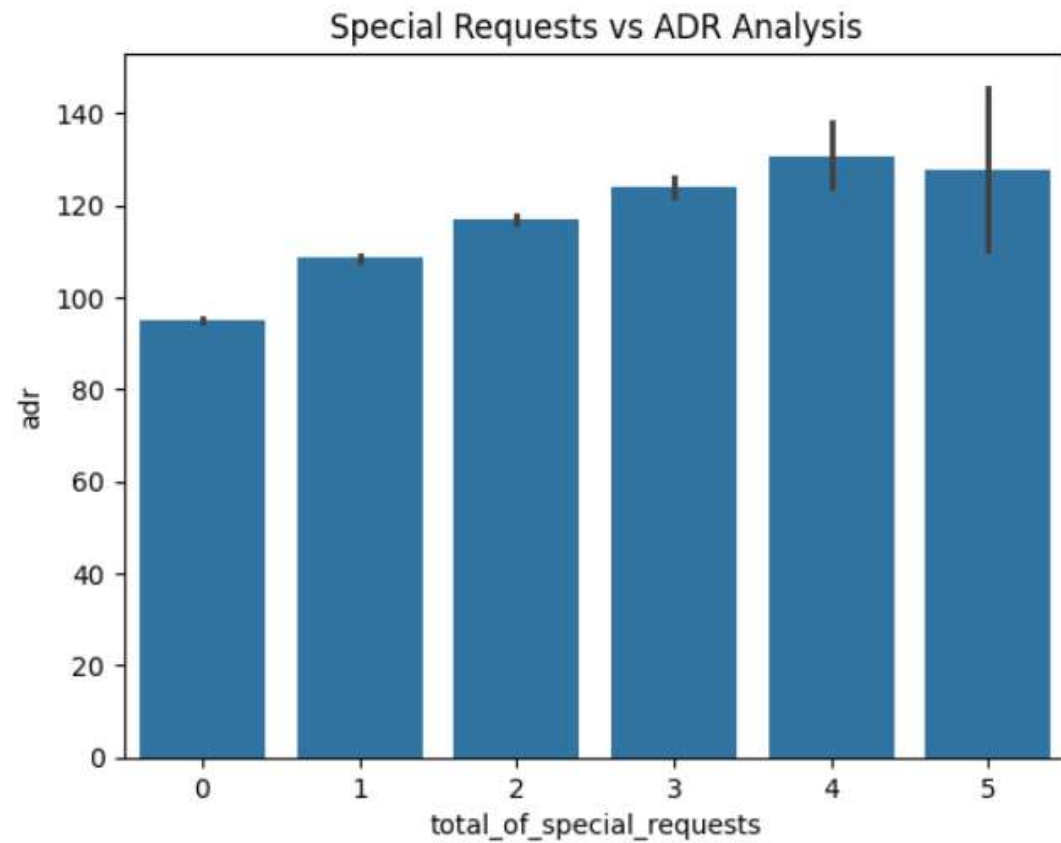


```
[13]: sns.barplot(  
      x='total_of_special_requests',  
      y='adr',  
      data=df  
      )  
  
plt.title('Special Requests vs ADR Analysis')  
  
plt.show()
```



```
[3]: import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df = pd.read_csv("Hotel_project.csv")

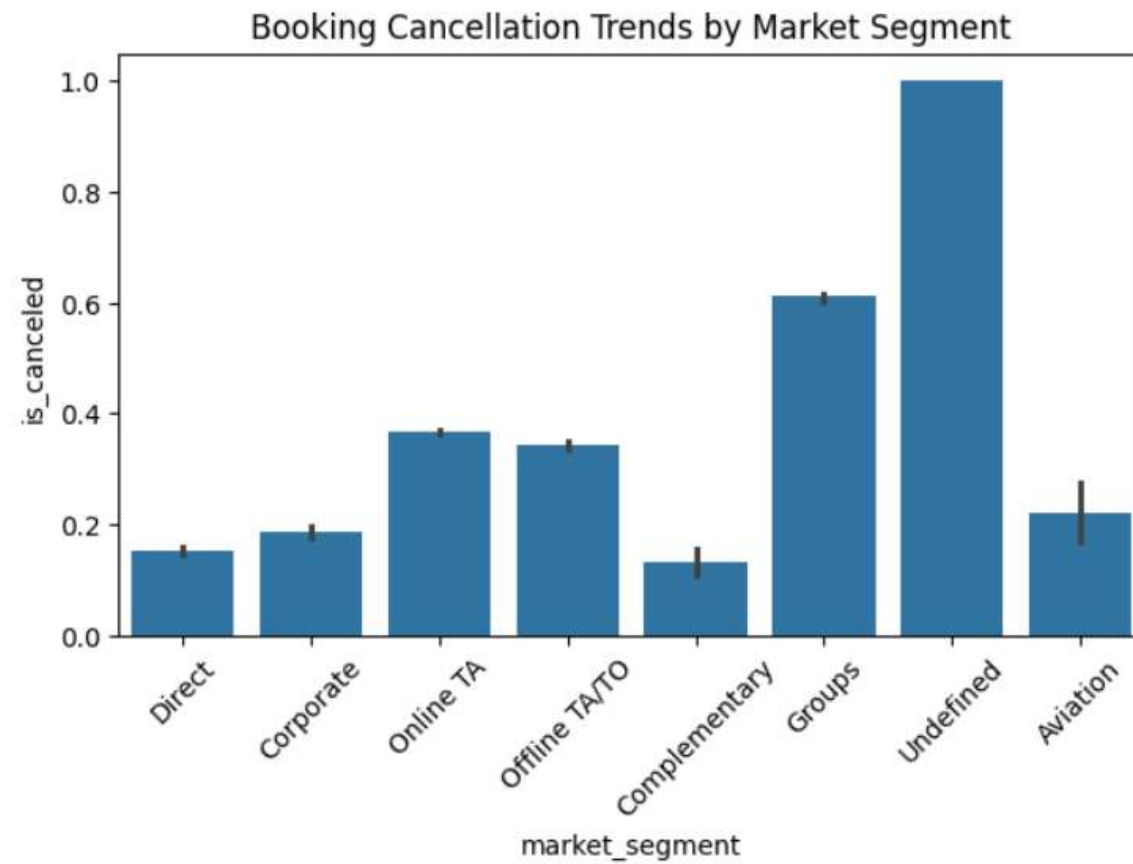
plt.figure(figsize=(7,4))

sns.barplot(
    x='market_segment',
    y='is_canceled',
    data=df
)

plt.title('Booking Cancellation Trends by Market Segment')

plt.xticks(rotation=45)

plt.show()
```



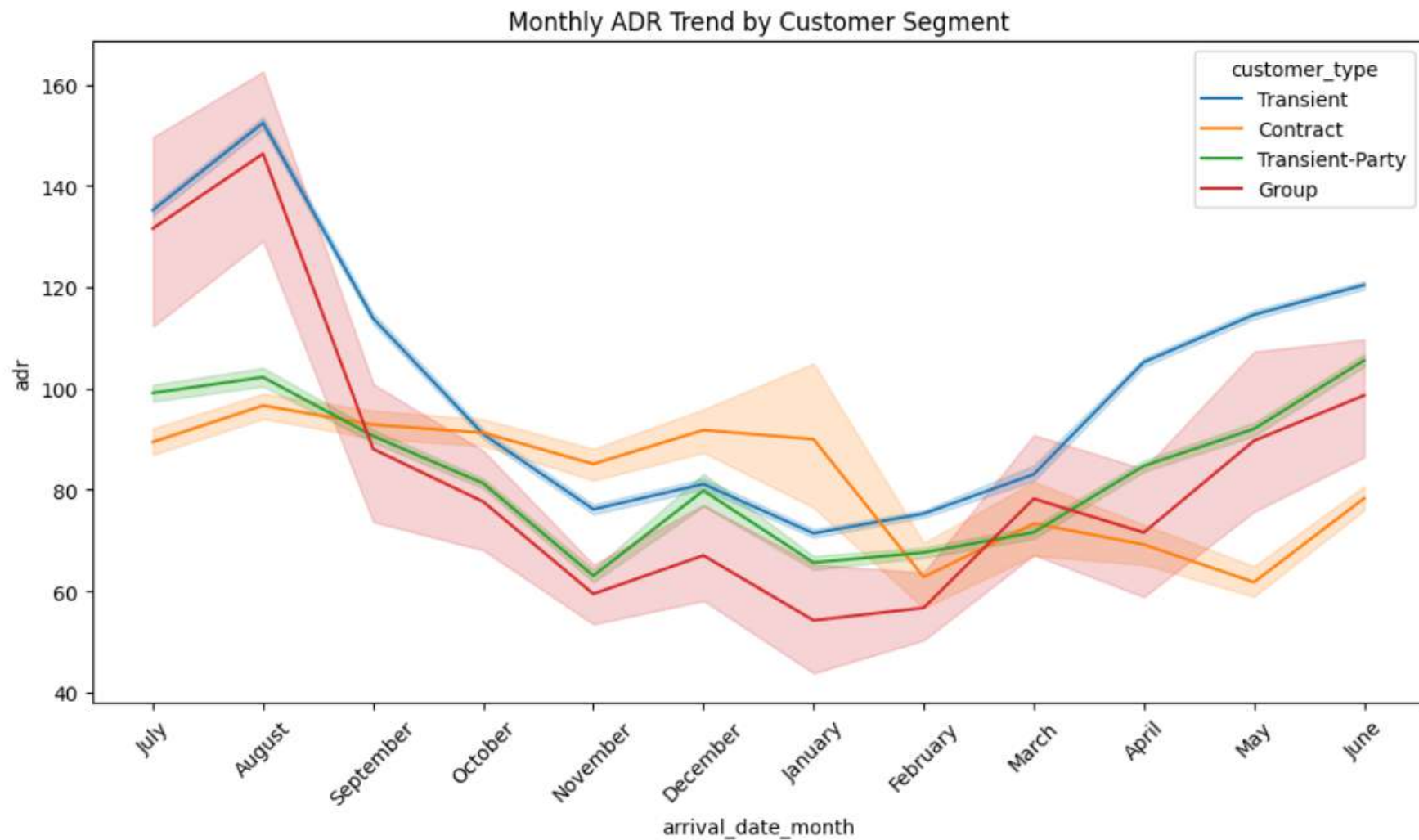
```
[11]: plt.figure(figsize=(12,6))

sns.lineplot(
    x='arrival_date_month',
    y='adr',
    hue='customer_type',
    data=df
)

plt.title('Monthly ADR Trend by Customer Segment')

plt.xticks(rotation=45)

plt.show()
```

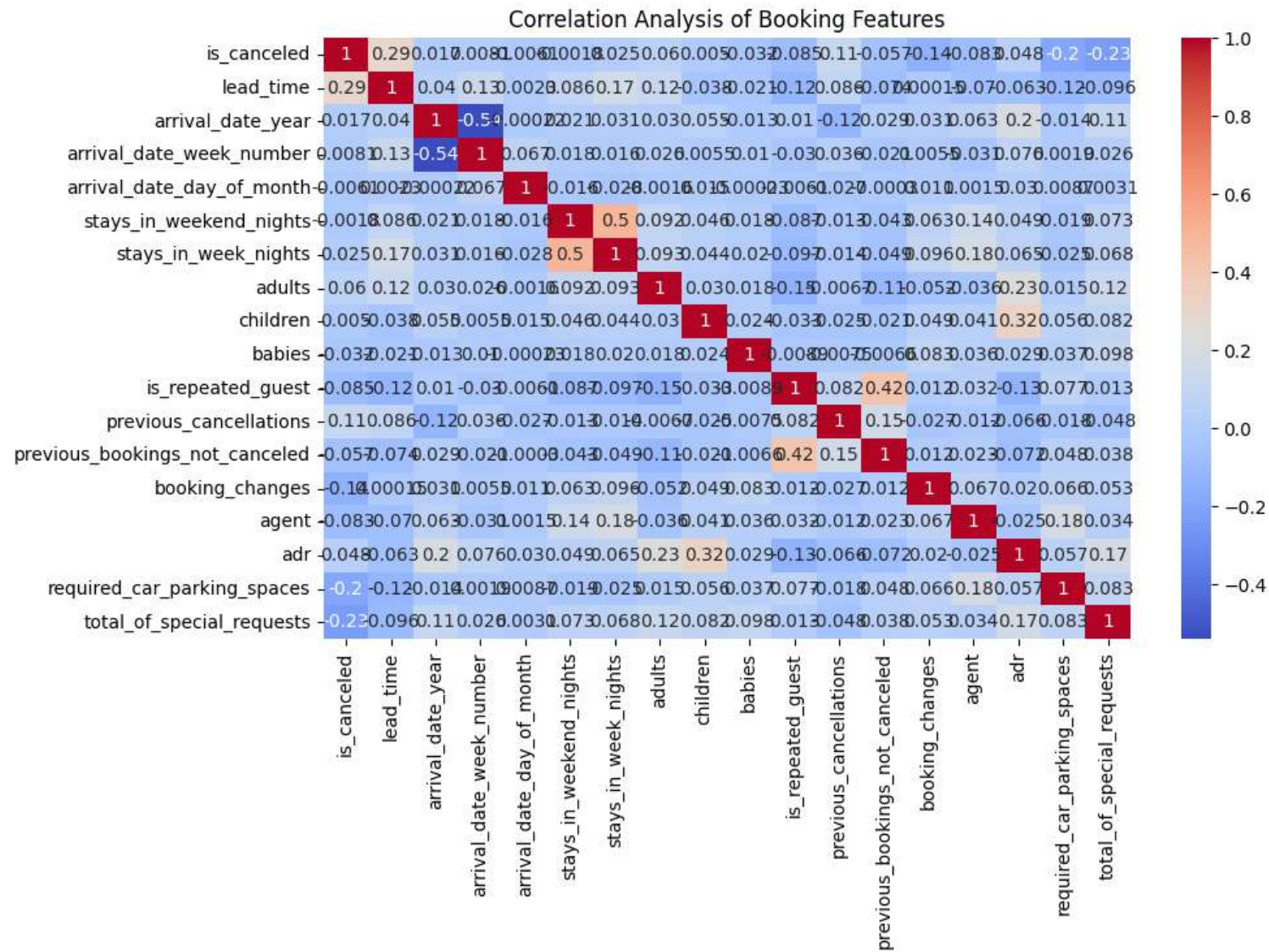


```
[16]: plt.figure(figsize=(10,6))
```

```
sns.heatmap(  
df.corr(numeric_only=True),  
annot=True,  
cmap='coolwarm'  
)
```

```
plt.title('Correlation Analysis of Booking Features')
```

```
plt.show()
```

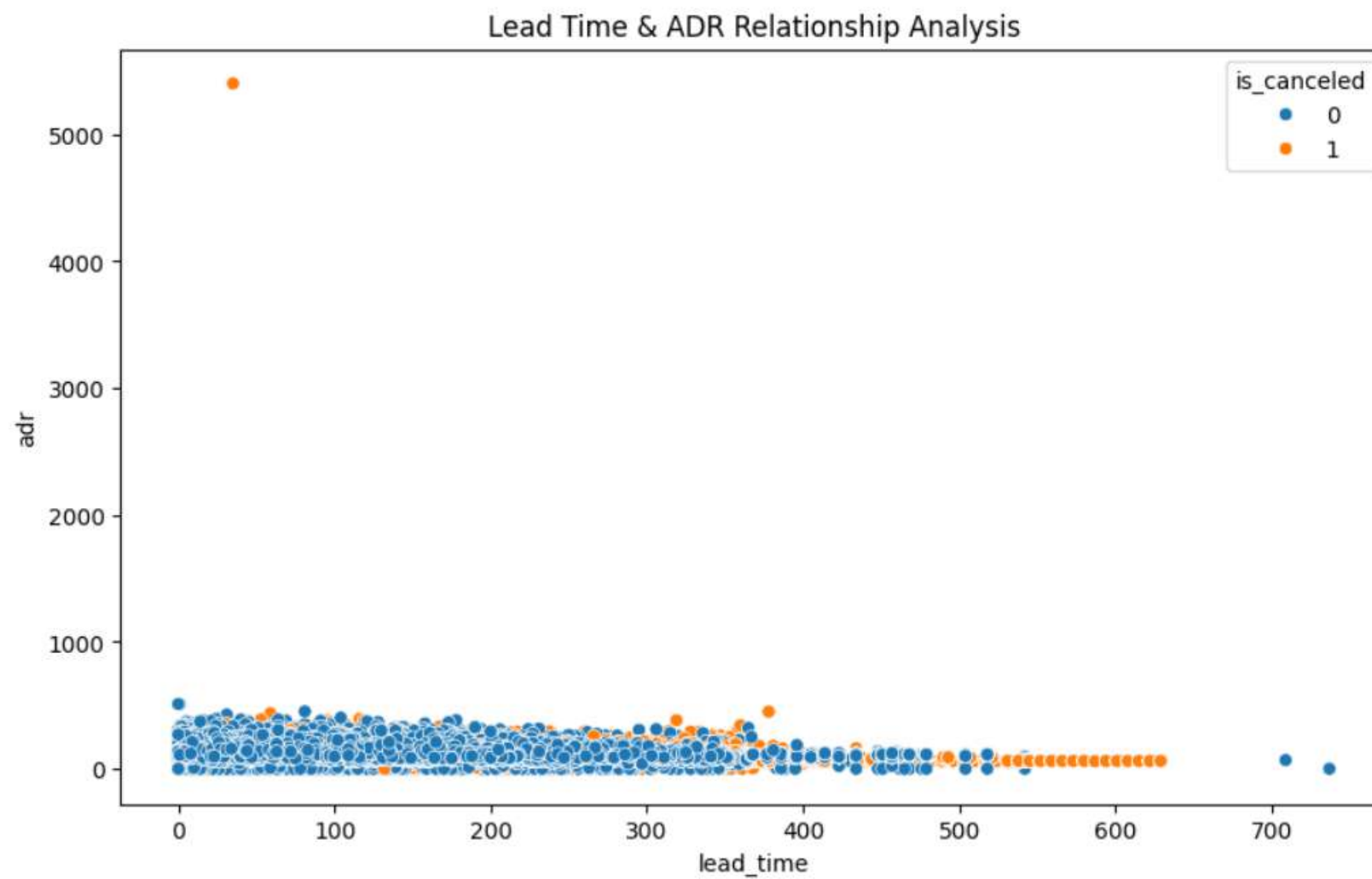


```
[8]: plt.figure(figsize=(10,6))
```

```
sns.scatterplot(  
    x='lead_time',  
    y='adr',  
    hue='is_canceled',  
    data=df  
)
```

```
plt.title('Lead Time & ADR Relationship Analysis')
```

```
plt.show()
```



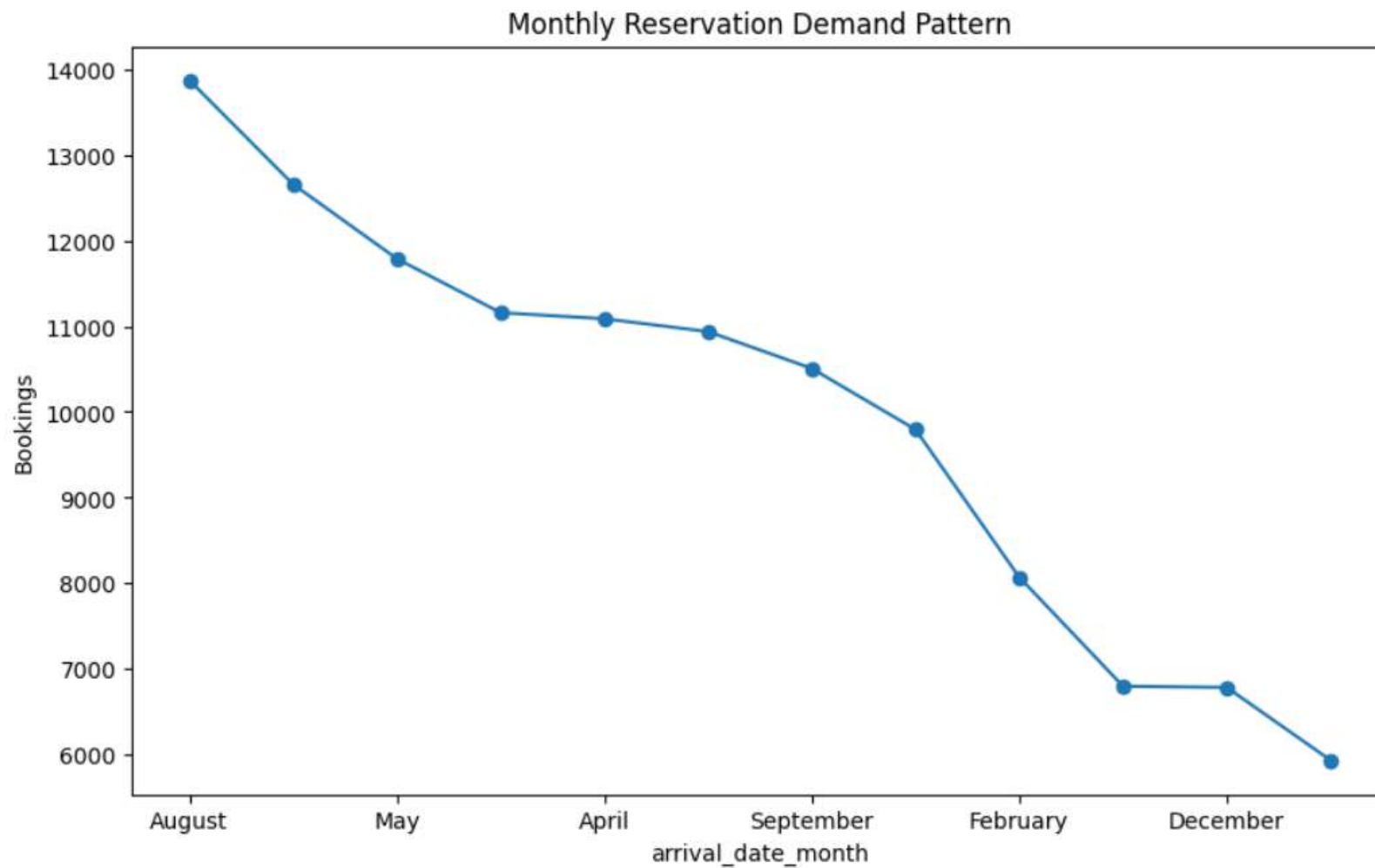
```
[5]: monthly = df['arrival_date_month'].value_counts()

monthly.plot(
    kind='line',
    figsize=(10,6),
    marker='o'
)

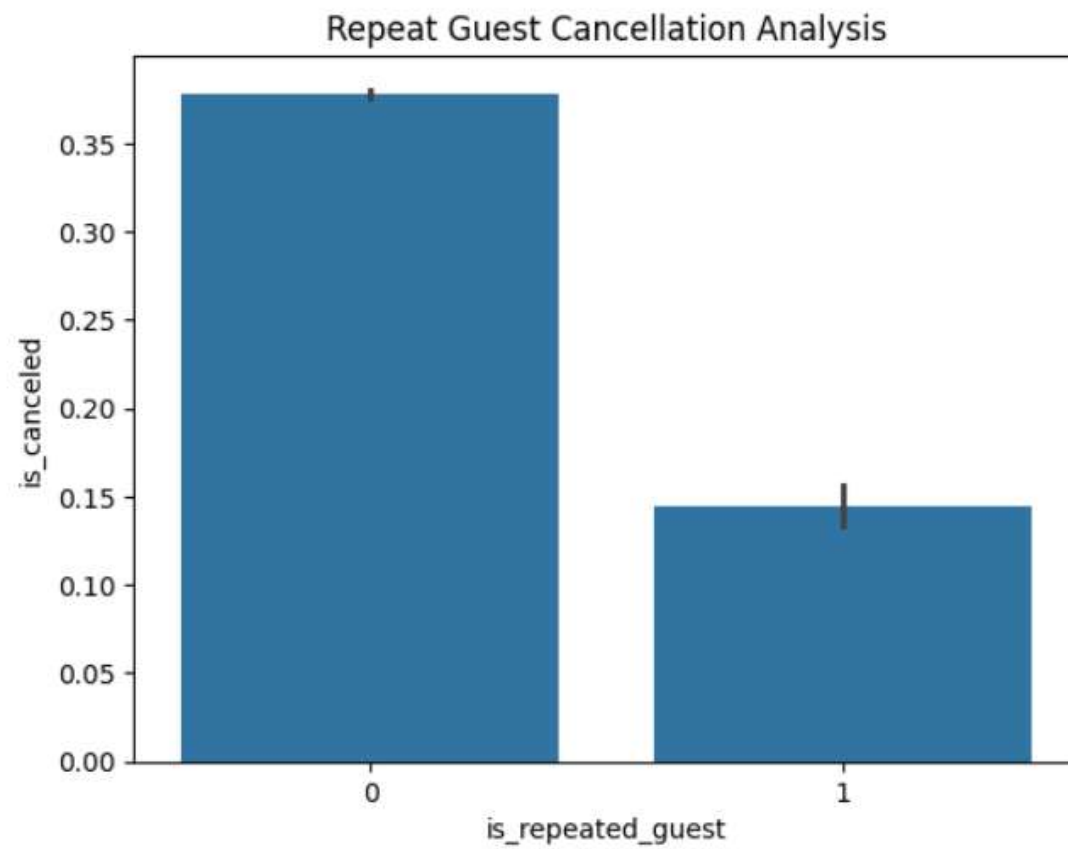
plt.title('Monthly Reservation Demand Pattern')

plt.ylabel('Bookings')

plt.show()
```




```
[14]: sns.barplot(  
x='is_repeated_guest',  
y='is_canceled',  
data=df  
)  
  
plt.title('Repeat Guest Cancellation Analysis')  
  
plt.show()
```

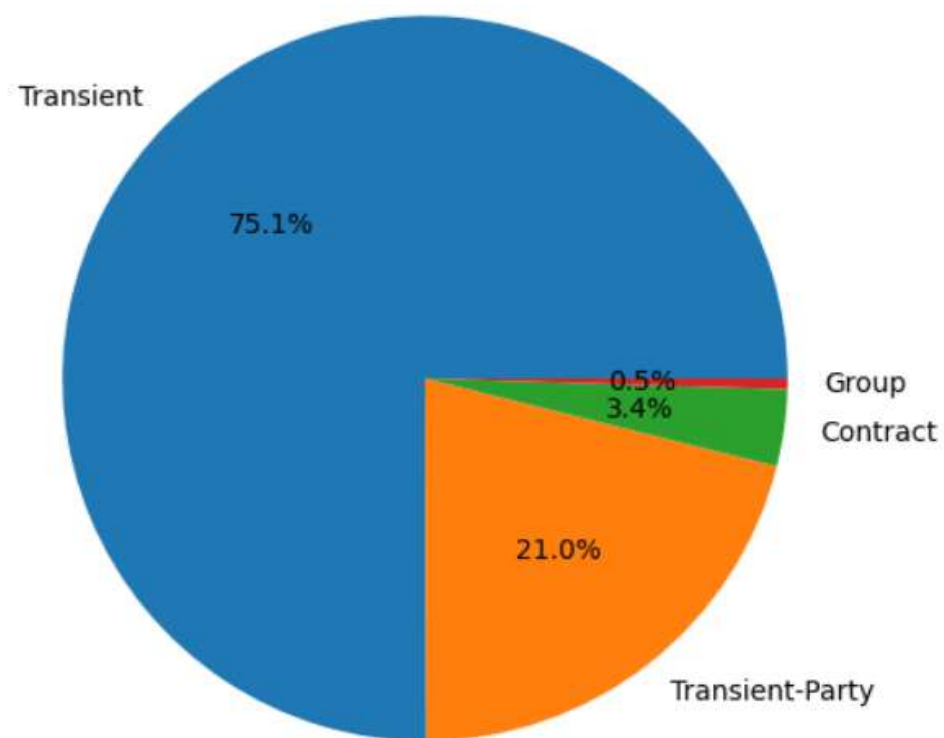


```
[ ]:
```

[15]:

```
df['customer_type'].value_counts().plot(  
kind='pie',  
autopct='%1.1f%%',  
figsize=(6,6)  
)  
  
plt.title('Customer Type Distribution Analysis')  
  
plt.ylabel('')  
  
plt.show()
```

Customer Type Distribution Analysis




```
[4]: plt.figure(figsize=(8,6))

sns.boxplot(
x='hotel',
y='adr',
data=df
)

plt.title('ADR Distribution Across Hotel Types')

plt.show()
```

